Akoizo Original Practice - Form 3 - Answer key

Division B | Boomilever | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Why do most teams aim the chain near 41 cm rather than 45 cm?

A. Less reach means a smaller moment and smaller member forces

B. The rules require exactly 41 cm

C. Longer reach lowers the score directly

D. The bucket hits the wall at 45 cm

Correct: A

2. (2 points) A bonus-style structure breaks at 11,000 g. How is it scored?

A. On 11,000 g with no bonus and no violation

B. On 11,000 g plus the bonus

C. Participation points only

D. Tier 2 with a construction violation

Correct: A

3. (2 points) What is the first tiebreaker if two teams have equal scores?

Unit: The test apparatus, setup, and loading procedure.
Worked-case data: Plan the six-minute window for a team that has practiced setup.
Separate investigation observations:
Run 1: Setup took 5 min; fast pour; failed at 9 kg with violent swing.
Run 2: Setup took 2 min; slow pour with stick tips; held 15 kg.
Inspection: Both structures were identical builds from the same wood batch.

A. Fastest setup

B. Highest load

C. Lowest structure mass

D. Closest load estimate

Correct: C

4. (2 points) A 30 cm x 0.3 cm x 0.3 cm balsa stick has a mass of 0.405 g. What is its density?

A. 1.5 g/cm^3

B. 0.45 g/cm^3

C. 0.15 g/cm^3

D. 0.015 g/cm^3

Correct: C

5. (4 points) Explain why the top chord can be much thinner than the compression strut.

answer: Wood is strong in tension along the grain and a tension member cannot buckle, while a slender strut buckles under compression at a much lower force. (4 points).

6. (4 points) Why is a reliable 7 g design better than a one-time 6 g success?

answer: Wood varies, so a design that works every time gives a dependable score, while a borderline design may fail at competition and score far lower. (4 points).

7. (4 points) Describe one way to keep the Loading Block centered within 2.5 cm of the wall centerline.

answer: Build two short rails or a small cradle at the loading point that fit the 5 cm block, so it cannot slide sideways off center. (4 points).

8. (6 points) Solve this case and explain your reasoning: Compare strut forces for a full 15 kg (147 N) at 42 cm reach, with contact at 20 cm versus 15 cm.

reasoning: At 20 cm: strut length sqrt (42^2 + 20^2) about 46.5 cm; force = 147 x 46.5 ÷ 20 about 342 N.
At 15 cm: length about 44.6 cm; force about 437 N.
Ratio: 1.28 times more compression at the shallower contact. (3 points).

conclusion: The bonus geometry asks the strut for more force over a similar length, so it must be stiffer or better braced. (3 points).

9. (2 points) Define “Structural efficiency” in the context of Boomilever.

answer: Load carried per gram of structure; here, the score itself. (2 points).

10. (4 points) How should a team choose its Estimated Load Supported?

Unit: The test apparatus, setup, and loading procedure.
Worked-case data: Plan the six-minute window for a team that has practiced setup.
Separate investigation observations:
Run 1: Setup took 5 min; fast pour; failed at 9 kg with violent swing.
Run 2: Setup took 2 min; slow pour with stick tips; held 15 kg.
Inspection: Both structures were identical builds from the same wood batch.

answer: Base it on consistent test data from identical builds; a team that reliably holds 15 kg should estimate 15,000 g. (4 points).

11. (6 points) Solve this case and explain your reasoning: A strut buckles at 120 N with an unbraced length of 30 cm. A brace is added at its midpoint. Estimate the new buckling load.

reasoning: Euler load scales with 1/L^2.
New unbraced length is 15 cm, half the original.
Load rises by (30/15)^2 = 4: about 480 N. (3 points).

conclusion: One midpoint brace roughly quadruples buckling resistance, usually for less mass than a thicker strut. (3 points).

12. (2 points) Define “Static equilibrium” in the context of Boomilever.

answer: Forces and moments both sum to zero, so nothing moves. (2 points).

13. (2 points) What should the team do next?

Supplied synthetic observations:
Builds 1-3: 8.1 g, 8.3 g, 8.2 g: all held 15 kg.
Builds 4-5: 7.4 g, 7.5 g: failed at 12 kg and 14 kg, strut buckled.
Build 6: 7.6 g with a stiffer, denser strut: held 15 kg.

A. Return to 8.2 g and stop testing

B. Repeat Build 6 to confirm it is reliable

C. Compete with Build 4 because it is lightest

Correct: B

14. (2 points) Define “Shear” in the context of Boomilever.

answer: Force that slides one surface along another. (2 points).

15. (2 points) Define “Hypotenuse” in the context of Boomilever.

answer: Strut length, sqrt (reach^2 + depth^2) in the triangle model. (2 points).

16. (2 points) Define “Force couple” in the context of Boomilever.

answer: Two equal, opposite, offset forces that balance a moment. (2 points).

17. (2 points) Define “Build log” in the context of Boomilever.

answer: Record of each build’s materials, mass, load, and failure. (2 points).

18. (2 points) Define “Stabilizing sticks” in the context of Boomilever.

answer: Dowels with doorstop tips, the only allowed way to steady the bucket. (2 points).